

DigiCert TLS in Kubernetes

What is DigiCert?

- DigiCert is a globally trusted Certificate Authority (CA) that provides SSL/TLS certificates.
- 👉 It provides publicly trusted SSL/TLS certificates
- 💡 Used for:
 - 🌐 Websites (HTTPS)
 - 🔗 APIs
 - 🌀 Kubernetes Ingress & Gateway API
 - 🕸 Service Mesh (Istio)
 - 📱 Apps & IoT

Why Use DigiCert?

👉 When you need trusted, production-grade security

🧩 Feature	💡 Why it Matters
🔑 Global Trust	🌐 Trusted by all browsers & OS → no security warnings
🏢 Enterprise Security	🛡 Strong validation ensures safe & compliant systems
⚡ Fast Issuance	🕒 Certificates can be issued quickly (DV in minutes)
📋 Multiple Options	🎯 Choose from DV , OV , EV , Wildcard , SAN certificates
🔄 Reliability	💪 Stable, trusted, and widely supported
Long Validity	⚡ Up to 1 year

Types of DigiCert Certificates

🧩 Type	📌 Description	🎯 Use Case
✅ DV (Domain Validation)	🌐 Verifies domain ownership only ⚡ (Fast & cheap)	📝 Blogs, personal & small sites

 Type	 Description	 Use Case
 OV (Organization Validation)	 Verifies domain + company	 Business websites & apps
 EV (Extended Validation)	  Highest trust level + Shows company name in browser	 Banking, e-commerce, finance
 Wildcard Certificate	 Covers all subdomains	 *.example.com (api.example.com, multiple subdomains)
 Multi-domain (SAN)	 Multiple domains in one cert	 Multi-app / (app.org , shop.net) multi-domain systems

How DigiCert Works (Trust Chain)

-  Trust Chain:
 -  Root CA (trusted globally)
 -  Intermediate CA
 -  Your certificate
 -  Browser verifies full chain → shows  padlock

How to Use DigiCert TLS in Kubernetes

-  Instead of self-signed or Let's Encrypt, you use DigiCert certs
-  Best for : Enterprise apps, Public production systems

Generate CSR (Certificate Signing Request)

```
openssl req -new -newkey rsa:2048 -nodes \
-keyout myapp.key \
-out myapp.csr \
-subj "/C=IN/ST=YourState/L=YourCity/O=YourOrg/CN=myapp.example.com"
```

-  Output:
 - myapp.key → Private key 
 - myapp.csr → Upload to DigiCert 

CSR Details

- Replace values:
 - C → Country (IN)
 - ST → State
 - L → City
 - O → Organization
 - CN → Domain name (Prasanthpoultry.com)



Buy Certificate from DigiCert

- 1 Go to DigiCert website
- 2 Choose certificate type
- 3 Upload CSR
- 4 Complete verification

- 👉 You will receive:
 - certificate.crt
 - intermediate.crt



Create Full Chain

```
cat certificate.crt intermediate.crt > fullchain.crt
```



Create Kubernetes TLS Secret

```
kubectl create secret tls myapp-tls \  
--cert=fullchain.crt \  
--key=myapp.key \  
--namespace=default
```



Use in Ingress

```
tls:  
- hosts:  
  - myapp.example.com  
secretName: myapp-tls
```

Access Your App

- 👉 <https://myapp.example.com>
 - ✅ Secured with DigiCert
 - ✅ Trusted by browsers
 - ✅ No security warnings

DigiCert vs Let's Encrypt vs Self-Signed

 Feature	 DigiCert	 Let's Encrypt	 Self-Signed
 Trust	✅ High (globally trusted)	✅ High (globally trusted)	⚠️ Not trusted by default
 Cost	💰 Paid	🆓 Free	🆓 Free
 Automation	⚠️ Limited / via tools	✅ Full (auto renew via ACME)	❌ Manual
 Validation Types	DV, OV, EV	DV only	None
 Validity	Longer (flexible)	Short (90 days)	Custom
 Use Case	 Enterprise, compliance	 Public websites	 Dev / internal systems

Summary

Important Notes

- 🔑 Keep private key safe
- 🔄 Renew before expiry
- 📄 Always include intermediate cert
- ❌ Don't expose raw cert files publicly

DigiCert provides:

- 🔑 High-trust certificates
-  Enterprise-grade security
-  Perfect for Kubernetes Ingress

-  Globally trusted HTTPS

E-commerce app:

-  Ingress → DigiCert TLS
-  Customers → HTTPS secure
-  Payment data protected
-  Trusted brand (no warnings)

 Best choice for production apps 